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(1) Research Program/Project/Study Title: ONLINE INFORMATION SYSTEM FOR STUDENT SERVICES OF CAPIZ STATE UNIVERSITY PONTEVEDRA	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	PONTEVEDRA
Province:	CAPIZ
Region:	VI – WESTERN VISAYAS

(5) Classification Research: Development: ☐ Basic ☒ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity EDUCATION SECTOR	(8) Discipline/Field of Specialization INFORMATION TECHNOLOGY
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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

This study aimed to develop an online information system for student services of Capiz State University Pontevedra. The system was designed to provide students with easy access to a variety of academic and administrative student related services, including the services for admission, guidance and counseling services, student organization, student handbook, student publication, complaint or incident report and other student services, all in one centralized location. The system was then developed using modern web technologies and tested with a small group of users to evaluate its effectiveness and identify areas for improvement.

The results of the study showed that the online information system was highly accepted in terms of the product quality and quality in use based on the ISO 25010 international standard for evaluation of software in providing students with improved accessibility to important student services. Students were able to access the services offered by the campus virtually in a more efficient and effective way, and the system helped to simplify administrative processes and increase communication between students, faculty, and staff. The study also revealed several areas for improvement, including the need for additional features and functionality and enhanced security measures. These findings will be used to guide future development of the system.

Overall, the development of an online information system for student services of Capiz State University – Pontevedra represents an important step forward in providing students with the support and resources they need to succeed academically and professionally. The system has the potential to enhance the reputation and competitiveness of the university, attract more students, and improve the overall quality of education provided.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

- Website
- Admission
- Counseling
- Booking
- GoDaddy
- Database

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

The increasing demand for quality higher education services has prompted many universities to implement technological advancements that can provide better access to information and services for their students. In response to this need, the researchers of this campus has initiated the development of an online information system for student services.

The online information system aims to provide students with a comprehensive and centralized platform that enables easy access to academic and administrative services such as services for admission, guidance and counseling, information & orientation, safety & security, student organization, student handbook, student publication, complaint or incident report and other student services. By providing these services through an online platform, the proponents aims to enhance accessibility, simplify processes, increase efficiency, and promote communication between students and university staff.

The system received positive feedback from users, including students, faculty, and staff, who appreciated the system's user-friendly interface, easy accessibility, and streamlined processes. One student mentioned that the system is "very convenient and saves a lot of time," while a faculty member noted that "it helps to reduce administrative burden and improve communication with students."

In summary, the development of an online information system for student services at Capiz State University-Pontevedra campus is a significant initiative that can provide numerous benefits to students and the university community. The reviews and feedback received from users suggest that the system is effective and efficient, and the university can use this feedback to guide future development and refinement of the system to further improve its effectiveness and user satisfaction.

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(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science research and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

MATERIALS

The materials to be used in conducting this study and in creating the online information system for student services of Capiz State University Pontevedra are as follows,

- Laptop or Personal Computer
- Mobile Phone
- Wi-Fi Router or Broadband or Data Connection

HARDWARE SPECIFICATIONS

Hardware	Minimum Requirements
Computer or Laptop	4 GHz minimum, multi-core processor
Mobile Phone	Android 8.0+
Memory (RAM)	At least 4GB
Hard Disk Space	At least 10 GB
Wi-Fi Router or Data Connection	At least 4G connectivity

SOFTWARE SPECIFICATIONS

Software	Definition
<i>For Local Development:</i>	
• Laragon	Local Server
• Visual Studio Code	Code Editor
• WordPress	Content Management Services (CMS)
<i>For Live Site:</i>	
• GoDaddy	Web Hosting and Domain Name Provider
• WordPress	Content Management Services (CMS)

BROWSER COMPATIBILITY

- Microsoft Edge
- Mozilla Firefox
- Google Chrome
- Safari

SOFTWARE DESIGN MODEL

Software Development Life Cycle (SDLC)

Iterative Development Model

In Iterative Model, iterative process starts with a simple implementation of a small set of the software requirements and iteratively enhances the involving version until the complete system is implemented and ready to be deployed. At each iteration, designed modifications are made and new functional capabilities are added. The basic idea behind these methods is to develop a system through repeated cycles and in smaller portion at a time. Following is the diagram for the software Development Life Cycle.

Below is the visualization of the Iterative Model to be used in this study.

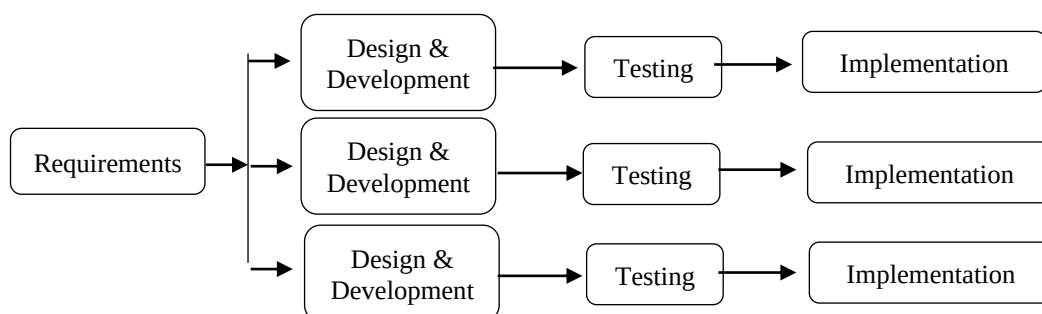


Figure 1. Iterative Model of the Online Information System for Student Services of Capiz State University Pontevedra

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SOFTWARE DEVELOPMENT PHASES

Requirement Gathering & Analysis: In this phase, requirements are gathered from customers and check by an analyst whether requirements will fulfil or not. Analyst checks that need will achieve within budget or not. After all of this, the software team skips to the next phase.

Design: In the design phase, team design the software by the different diagrams like Data Flow diagram, activity diagram, class diagram, state transition diagram, etc.

Testing: After completing the coding phase, software testing starts using different test methods. There are many test methods, but the most common are white box, black box, and grey box test methods.

Implementation: In the implementation, requirements are written in the coding language and transformed into computer programs which are called Software.

Deployment: After completing all the phases, software is deployed to its work environment.

Maintenance: The software is monitored to ensure it continues to function as it was designed to, and repairs or upgrades are performed as needed. The maintenance phase occurs once the software is fully operational.

CONCEPTUAL FRAMEWORK

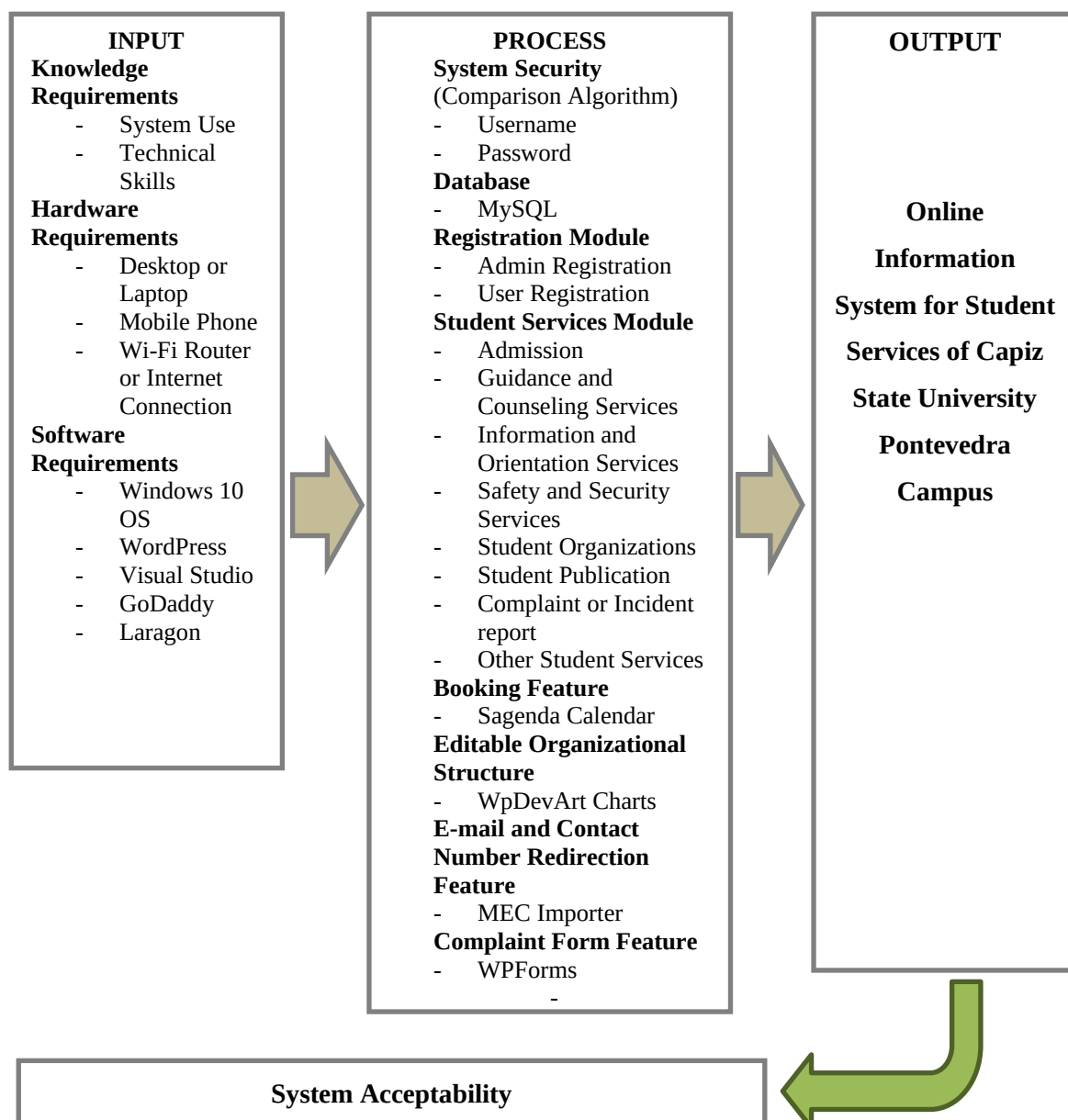


Figure 2. Conceptual Framework (IPO) of the Online Information System for Student Services of Capiz State University Pontevedra

SYSTEM ARCHITECTURE

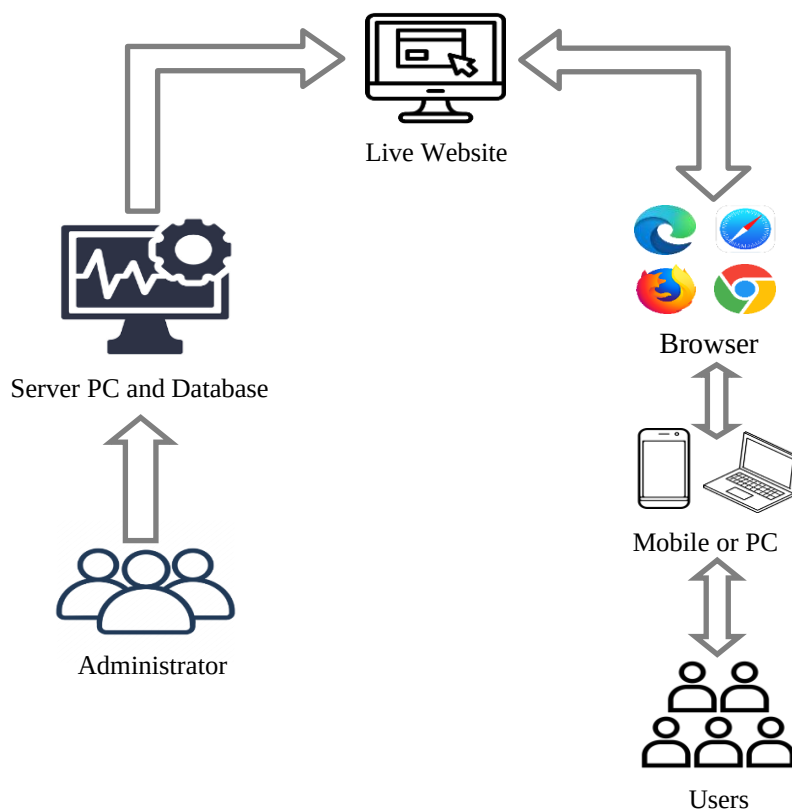


Figure 3. System Architecture of the Online Information System for Student Services of Capiz State University Pontevedra

INSTRUMENTS

The system was tested for final evaluation or acceptability by the end users using the ISO 920. It was evaluated in terms of product quality such as functional suitability, performance efficiency, compatability, usability, reliability, security, maintainability and portability, and quality in use such as effectiveness, efficiency, satisfaction, freedom from risk and context coverage.

The acceptability of the Online Information System for Student Services of Capiz State University Pontevedra were rated by the respondents using the scoring below:

Range of Score	Verbal Interpretation	Description
4.5 – 5.0	Highly Accepted	Respondent agrees to all the features of the system.
3.5 – 4.49	Accepted	Respondent agrees to about 80% of the system's features.
2.5 – 3.49	Moderately Accepted	Respondent is satisfied with the features it offers.
1.5 – 2.49	Slightly Accepted	Respondent is slightly beginning to dislike some of its features.
1.0 – 1.49	Unaccepted	Respondent doesn't accept all the features of the system

STATISTICAL TOOLS

A total of forty (50) respondents was sampled in the study using a simple random sampling method. These respondents viewed and analyzed the website and were given the questionnaire for them to be answered. To make data analysis easier, a tally sheet was provided where the responses of the respondents were recorded.

To avoid repetition of encoding the responses, the questionnaires were consecutively numbered. The gathered data were tabulated and prepared for analysis. The statistical measure used for the acceptability is the mean and standard deviation.

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Mean. The statistical tool that measures central tendency as the sum of all scores divided by the number of cases. In this study it was used to calculate the average central tendencies of any group of responses. The formula for solving the mean was:

$$\bar{x} = \frac{\sum x}{n}$$

where: $\sum x$ = the sum of the scores
 n = number of respondents

Standard Deviation. A statistical tool that measures the dispersion of a data set relative to its mean and is calculated as the square root of variance. The formula for solving the standard deviation was:

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

n = The number of data points.

x_i = Each of the values of the data.

$\bar{x}$ = The mean of x_i .

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Below are the discussions on the acceptability of the website or online information system in terms of product quality and quality in use.

PRODUCT QUALITY

The overall results of Online Information System for Student Services of Capiz State University Pontevedra in terms of all aspects in product quality consisted of functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability got a grand mean of 4.56 which can be interpreted as *Highly Accepted*, meaning the respondents agreed in terms of product quality of the system. On the other hand, the standard deviation for the result of Product Quality is .03919.

Product Quality	Mean Average	Verbal Interpretation
1. Functional Suitability	4.55	Highly Accepted
2. Performance Efficiency	4.5	Highly Accepted
3. Compatibility	4.55	Highly Accepted
4. Usability	4.55	Highly Accepted
5. Reliability	4.62	Highly Accepted
6. Security	4.54	Highly Accepted
7. Maintainability	4.54	Highly Accepted
8. Portability	4.61	Highly Accepted
GRAND MEAN	4.56	Highly Accepted

Table 1. Summary of Results in Product Quality

QUALITY IN USE

The quality in use consisted of effectiveness, efficiency, satisfaction, freedom from risk and context coverage got a grand mean of 4.63 which can be interpreted as *Highly Accepted*. Meaning the respondents agreed in terms of quality in use of the system. Conversely, the standard deviation for the result of Quality in Use is .04615.

Product Quality	Mean Average	Verbal Interpretation
1. Effectiveness	4.7	Highly Accepted
2. Efficiency	4.65	Highly Accepted
3. Satisfaction	4.59	Highly Accepted
4. Freedom from Risk	4.59	Highly Accepted
5. Context Coverage	4.64	Highly Accepted
GRAND MEAN	4.63	Highly Accepted

Table 2. Summary of Results in Quality in Use

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(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

FINDINGS

The designed and developed system was called Online Information System for Students Services on the gathered data, the following findings were made:

1. The Online Information System for Student Services of Capiz State University Pontevedra was designed, developed and tested for acceptability.
2. The modules for Admission, Guidance and Counseling Services, Information and Orientation Services, Safety & Security Services, Student Organizations, Student Handbook, Student Publication, Complaint or Incident report, and Other Student Services of the online information system for student services was designed and develop properly.
3. The developed online information system for student services achieved the following features for System Security, Database, Scheduling, Editable Organizational Structure, Complaint Form, E-mail and Contact Number Redirection and Report Generation Feature.
4. The overall results of Online Information System for Student Services of Capiz State University Pontevedra in terms of all aspects in product quality which consisted of sub-categories which are functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability got a grand mean of 4.56 which can be interpreted as “Highly Accepted”. On the other hand, the level of acceptability in quality in use which consist of sub-categories namely effectiveness, efficiency, satisfaction, freedom from risk and context coverage got a grand mean of 4.63 and can be interpreted as “Highly Accepted” meaning the respondents agreed in terms of product quality and quality in use of the system.

CONCLUSION

The study confirmed that the Online Information System for Student Services of Capiz State University Pontevedra was designed, developed, and tested successfully. The system’s modules for different student services were properly implemented, and it received high acceptance ratings for both product quality and quality in use demonstrating its effectiveness, reliability, and usability in providing a range of student services. Overall, the successful implementation of this system can greatly enhance the efficiency and convenience of accessing and managing student-related information and services at the university.

RECOMMENDATIONS

Based on the development and implementation of the Online Information System for Student Services of Capiz State University Pontevedra, the following recommendations were offered:

1. Continuous improvement and development of the Online Information System for Student Services should be prioritized, to ensure that it remains up-to-date and relevant to the needs of the students and the campus.
2. Regular training and support should be provided to both students and staff on how to use the system effectively, to maximize its benefits.
3. Additional features and services can be added to the system to further improve the student experience and increase the system’s functionality, such as online payment, and a feedback system.
4. Further research and evaluation should be conducted to measure the long-term impact of the Online Information System for Student Services on student satisfaction, academic performance, and overall campus or university operations.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

The output of the Online Information System for Student Services of Capiz State University Pontevedra is the website or web-based platform that provides a range of services and information to students. The system aimed to enhance the delivery of student services and support, improve communication channels between students and the university, and simplify administrative processes.

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1. A user-friendly interface that allows students to easily access information and services related to admission, guidance and counseling, information & orientation, safety & security, student organization, student handbook, student publication, complaint or incident report, and other student services.
2. A centralized database that stores and organizes relevant information and services for students, enabling them to access the resources quickly and easily they need.
3. A secure login system that ensures the confidentiality and privacy of student information, while also enabling students to access personalized services and information based on their individual needs and preferences.
4. An admission services that allow students to apply for admission online. The system will also provide information on the admission process, requirements, admission officer contact details, etc.
5. A guidance and counseling services that will offer online counseling services to students, allowing them to schedule appointments with counselors and access resources on mental health and well-being.
6. An information and orientation services that will provide students with information on university policies, procedures, and regulations. It will also offer online orientation sessions to new students, providing them with an overview of university services and resources.
7. A safety and security services that will provide students with information on campus safety and security measures. Students will be able to access emergency resources.
8. A student organizations module that will allow students to access the resources of different student organizations that is duly recognized by the campus via online. It will provide information on the different organizations and their objectives, constitution and by-laws, officers, social media platform, and etc.
9. The student handbook that will provide access to the university's student handbook, which contains information on academic policies, procedures, and regulations.

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

A technology transfer activity design of the Online Information System for Student Services of Capiz State University Pontevedra as an extension activity of the campus could be designed as follows:

1. Development of the Online Information System: Based on the identified needs, design and develop the Online Information System for Student Services. This could involve hiring IT professionals and web developers to enhance the website and the necessary features.
2. Training and capacity building: Once the Online Information System has been developed, conduct training and capacity-building activities for university staff, including the IT team, department heads, and front-line staff. This will ensure that they have the necessary skills to operate and maintain the system.
3. Pilot testing: Conduct a pilot test of the Online Information System to identify any glitches or issues that need to be addressed. This can be done by selecting a small group of students to use the system and provide feedback.
4. Launch and dissemination: Once the Online Information System has been fully developed and tested, launch it to the entire university community. This can be done through an official launch event, promotional materials, and targeted communication to different departments and units.
5. Maintenance and updates: Ensure that the Online Information System is regularly maintained and updated to keep it running smoothly and efficiently. This will involve regular checks for bugs, updates to security features, and improvements to functionality based on user feedback.
6. Evaluation and feedback: Finally, conduct an evaluation of the Online Information System to assess its impact and effectiveness. This can be done through surveys, focus group discussions, and usage data analysis. Use this feedback to improve the system further and to inform future technology transfer activities.

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(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	1	₱1,000.00
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
o Website Design and Development	-	₱5,000.00
o Web Hosting (12 months)	1	₱2,388.00
o Domain Name (12 months)	1	₱260.00
o Paid Extensions (12 months)	1	₱999.00
▪ Wpdevar Charts		
o Printing	-	₱500.00
o Mobile Internet/Data Load	-	₱2,000.00
Total		₱12,147.00

Table 3. Budgetary Requirements for designing the Online Information System

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

The researchers would like to express their deepest gratitude to their thesis adviser for her invaluable guidance, encouragement, and support throughout their thesis journey. Her expertise and constructive feedback have been instrumental in shaping the researchers and honing their skills. They are grateful for her unwavering commitment to excellence, which had motivated them to strive for excellence in their work.

They are grateful to their family and friends for their firm love, support, and understanding throughout their thesis journey. Their encouragement, patience, and belief in the researchers have been a source of strength and motivation. They would not have been able to achieve this milestone without them.

The researchers were also thankful to God for giving them the wisdom, strength, and perseverance to complete this thesis. Without His guidance, they would not have been able to overcome the challenges and obstacles that they faced throughout this journey.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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